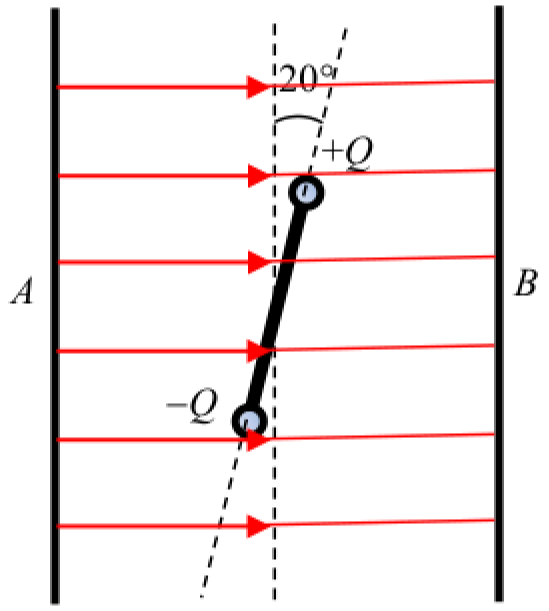
8.	(a)	(i)	- the air loses its insulating properties	1A		
			Or electrons or ions can pass through the air (between clouds and Earth or between clouds and clouds)	1A		
					1	
		(ii)	$E = \frac{V}{d}$	1M		
			$V = E\,d = (3 \times 10^5) \times 2000$			
			$= 6 \times 10^8$ V	1A		
					2	
	(b)	(i)	magnetic field into paper (due to upward lightning current)	1A		
			$B = \frac{\mu_0 I}{2\pi r}$	1M		
			$= \frac{4\pi \times 10^{-7} \times 30000}{2\pi \times 1500}$			
			$= 4 \times 10^{-6}$ T	1A		
					3	
		(ii)	When the lightning current is increasing, the induced current flows in the anticlockwise direction so as	1A		
			to oppose the increasing magnetic field (into paper).	1A		
			After reaching maximum, the lightning current is decreasing, the induced current flows in the clockwise / opposite direction.	1A		
					3	
		(iii)	Electric field (in the atmosphere).	1A		
			E-field increases / builds up (to threshold) before lightning happens	1A		
			Or Lightning current and magnetic field only exist during lightning.	1A		
					2	

<table>
<tr><td>9. (a) Top view

</td><td>
1A

1A

2</td><td>Accept: field lines are not on Figure 9.2
Correct direction (From A to B)
Accept: field lines not in contact with the plates
Perpendicular to plates, parallel and evenly spaced
Accept: 3 or more field lines
NOT accept: 2 or more mistakes in the drawing</td></tr>
<tr><td>(b) (i) $F \times d = (2.0 \times 10^{-5})(0.05 \cos 20°)$
$= 9.396926 \times 10^{-7}$ N m $\approx 9.40 \times 10^{-7}$ N m</td><td>1M
1A

2</td><td>1M $F \times d$
Accept: without consider the component / without component
d (0.05, 0.05/2, 2×0.05, 5, 5/2, 2×5)
NOT accept: calculation using separation between plates
d (0.1, 10)</td></tr>
<tr><td>(ii) $E = \frac{V}{d}$
$= \frac{5.0 \times 10^3}{0.1}$
$= 50\ 000$ V m^{-1} or N C^{-1} = 50 kV m^{-1} or kN C^{-1}</td><td>1M
1A
2</td><td>Accept: $\frac{5\times10^3}{0.1}, \frac{5\times10^3}{1}, \frac{5}{1}, \frac{5}{0.1}$</td></tr>
<tr><td>(iii) $E = \frac{F}{Q}$
$Q = \frac{F}{E} = \frac{2.0 \times 10^{-5}}{5.0 \times 10^4}$
$= 4.0 \times 10^{-10}$ C</td><td>1M
1A
2</td><td>e.c.f. from (b)(ii)</td></tr>
</table>